Week 3: Churn Analysis Report

By

TEAM 6

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1. Introduction

Objective

The primary objective of this churn analysis is to identify the underlying patterns, behavioral triggers, and demographic factors that lead to student disengagement within the Opportunity ecosystem. By examining the transition from "SignUp" to "Rewarded" or "Engaged" status, this analysis aims to pinpoint exactly where students fall out of the funnel and why. Specifically, we seek to quantify the impact of administrative delays, program duration, and student demographics on the **Risk_Outcome** variable.

Relevance to Improving Student Retention

Improving student retention is critical for maximizing the social and educational impact of the offered opportunities. This analysis provides actionable insights that allow for:

- **Proactive Intervention:** By identifying "At Risk" students early based on their Signup_Entry_Lag or Signup_Apply_Delay, program coordinators can trigger automated reminders or personalized support to keep the student on track.
- **Process Optimization:** Understanding how administrative bottlenecks (e.g., the time between signup and entry creation) correlate with churn allows the organization to streamline onboarding and reduce initial friction.
- **Program Design:** Insights into "Opportunity Duration Fatigue" can inform the development of shorter, high-impact milestones or "Events" that maintain student momentum more effectively than multi-year courses.
- **Resource Allocation:** By focusing retention efforts on the most significant predictors of churn, resources can be directed toward the regions and student groups where they will have the greatest impact on overall success rates.

2. Data Preparation

Data Cleaning notebook 1: Click here
Data Cleaning notebook 2: Click here

Cleaning Steps:

i. The columns' names were formatted in snake_case for readability and as a naming convention standard for a programming language like Python and SQL. "Instition Name" is now formatted as "Institution_Name"

ii. All date columns were converted to the correct data type using Python (pd.to_datetime).

iii. Missing values were handled using a column-specific approach to preserve data integrity. Categorical fields such as names and institutions were filled with neutral placeholders like "Unknown" or "Not Specified". Numerical and time-based features used sentinel values (–1) to explicitly represent missing or inapplicable cases without removing records. Where date imputation was necessary, missing dates were filled with a clearly identifiable sentinel date (1900-01-01) to distinguish them from valid timestamps. Sensitive attributes, such as date of birth, were not imputed to avoid introducing assumptions.

Data Quality Before and After Table:

Metric	Raw Data	Cleaned Data
Row Counts	**8558**	**8263**
Column Counts	**16**	**27**
Date Consistency	**Formatted as Object**	**Formatted Uniformly as Datetime**
Missing Value Counts	**3804**	**0**

Feature Engineering:

Feature Overview

1. Applied Flag

Did the learner apply after signing up?

1 = Yes, 0 = No

Shows real intent, not just signups.

2. Signup → Apply Delay (Days)

How long it took to apply after signup.

-1 = Never applied

Measures learner motivation.

3. Opportunity Duration (Days)

How long the opportunity lasts.

-1 = Ongoing or unknown

Helps compare short vs long programs.

4. Ongoing Opportunity Flag

Is the opportunity currently active?

1 = Yes, 0 = No

5. Signup Entry Lag (Days)

Delay between signup and system entry.

-1 = Not available

Indicates system/data delay, not learner behavior.

6. Learner Age

Age calculated from date of birth.

Blank = Not provided

7. Age Group

Age ranges (Student, Early Career, etc.).

Easier than using exact age.

8. Completed Flag

Did the learner meaningfully engage?

1 = Started/allocated/rewarded

0 = Did not engage

8. Risk Outcome

Overall learner status:

Engaged, Rewarded, Dropout, Rejected, Pending

Helps identify risk and success.

9. Region

Countries are grouped into regions (Asia, Africa, etc.).

Simplifies geographic analysis.

10. Signup Month & Year

When the learner signed up.

Helps track trends over time.

3. Exploratory Data Analysis (EDA)

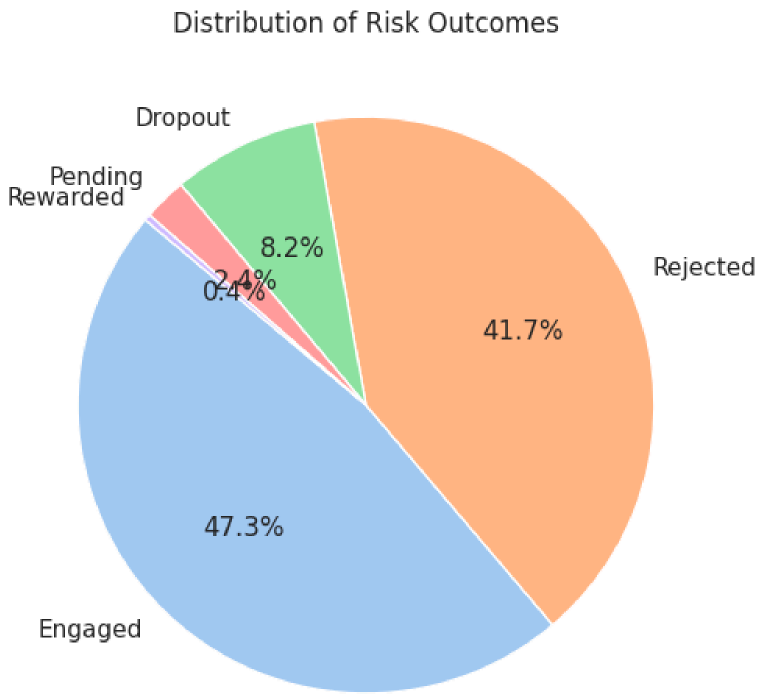
- Data visualization Link: [Click here]
- Descriptive Statistics: Key statistics of the dataset.

  The dataset captures a diverse range of learners, primarily in their early careers.

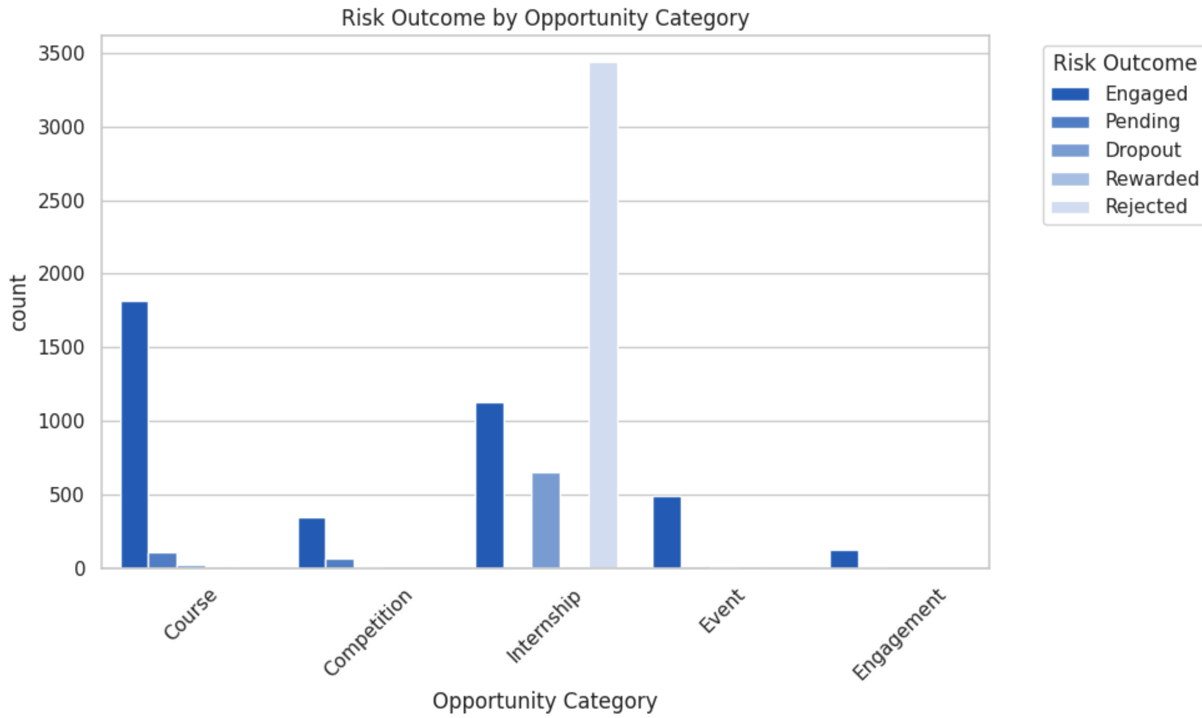
Metric	Learner Age	Opportunity Duration (Days)	Signup-Entry Lag (Days)	Signup-Apply Delay (Days)
Mean	25.80	187.18	382.45	77.37
Median (50%)	25.00	-1.00*	410.00	0.00
Std Dev	4.35	324.44	150.64	125.60
Min	7.00	-1.00	-1.00	-1.00
Max	59.00	841.00	669.00	695.00

*Note: Values of -1 in duration indicate ongoing opportunities without a fixed end date.

- Visualization:

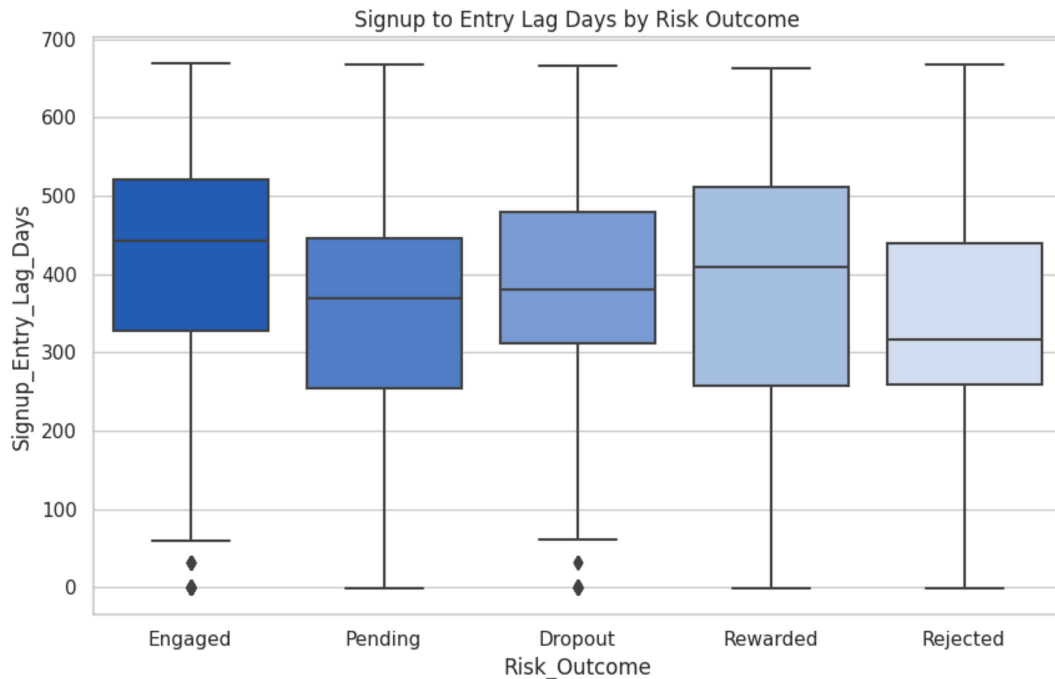


Risk Outcome Distribution The dataset shows that while **47.3%** of students are **Engaged**, a massive **41.7%** fall into the **Rejected** category, and **8.2%** are explicitly marked as **Dropout**. This indicates that the primary "churn" occurs not just from students leaving, but from high barrier-to-entry (rejections).

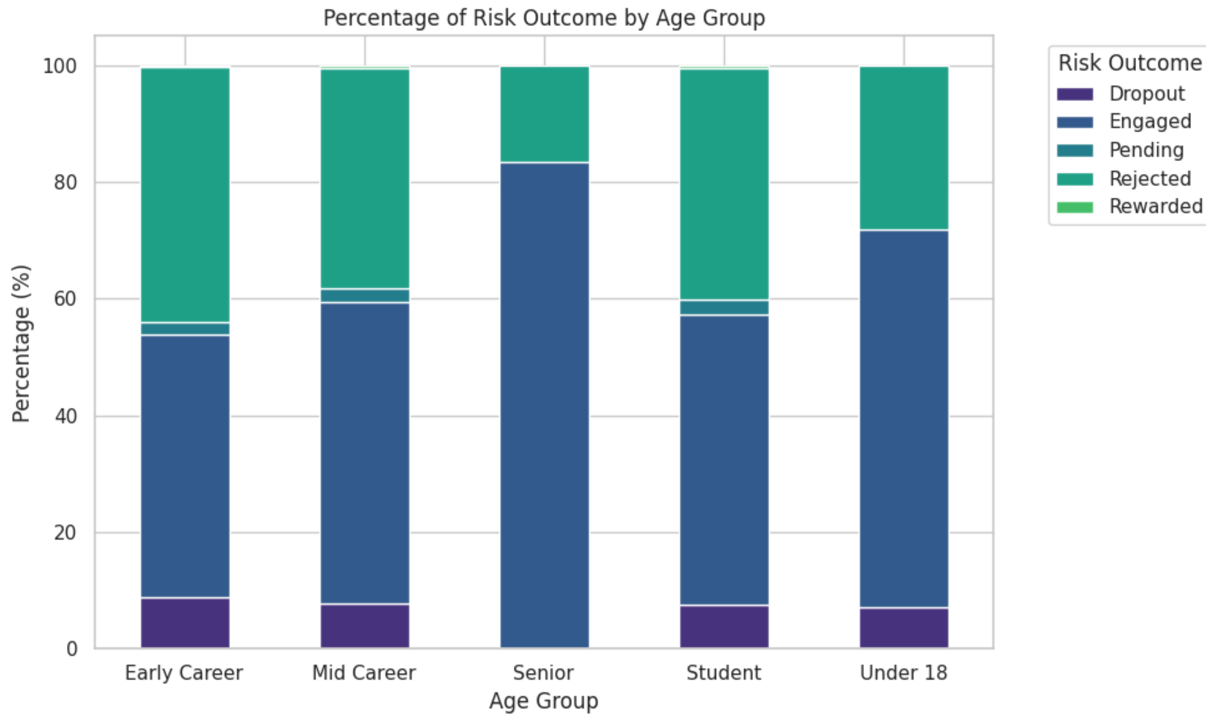


Risk by Opportunity Category * Courses: These attract the highest volume but also see the highest number of Rejections and Dropouts.

- **Events/Workshops:** Show a much higher ratio of "Engaged" status relative to their size, suggesting that short-term, high-intensity formats are better for retention.



Signup-to-Entry Lag vs. Risk The boxplot reveals that students who are eventually **Engaged** often have a wide range of lag times, but **Dropouts** and **Rejected** students often cluster around specific administrative windows. A lag of over **400 days** is remarkably common, suggesting that long-term "backlog" entries are a significant risk factor.



Risk by Age Group * Early Career (25-34): This group shows the most stability.

- **Students (18-24):** This group has a higher propensity for being "Rejected" or "Pending," likely due to stricter eligibility checks or fluctuating availability.

- Patterns and Trends:

  The "Delay" Correlation: There is a notable trend where the longer the `Signup_Apply_Delay_Days_Filled`, the higher the risk of rejection or dropout. Students who apply immediately (0-day delay) are significantly more likely to reach "Engaged" status.

  Institutional Clusters: Institutions like *Saint Louis University* drive the majority of the volume. Because of this, the overall dataset trends are heavily influenced by the administrative timelines of these large-scale partners.

High-Volume Rejection: The high rejection rate (41.7%) suggests that "Churn" in this ecosystem is often involuntary. Students are signing up for opportunities for which they may not qualify, or they are being timed out due to the high `Signup_Entry_Lag`.

Age Maturity: Older learners (35+) have lower representation but tend to have higher engagement rates, suggesting that "Professional" categories may have clearer intent than the general "Student" population.

4. Predictive Modeling

Objective:

We aimed to predict **Risk_Outcome** (Engaged vs At-Risk), as it represents the final engagement status and is actionable.

Model Workflow

- **Prepared the Data**
- Separated target (Risk_Outcome) and features
- Converted all categorical and boolean columns into numeric form
- Removed date-type columns to avoid model errors
- **Split the Dataset**
- Used an **80% training and 20% testing split** with stratification to maintain class balance.
- **Selected the Model**
- Chose **Random Forest** because it handles non-linear patterns, reduces overfitting, and works well with mixed features.
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- **Trained the Model:** Trained the Random Forest on the training data using multiple decision trees. Link: click here
- **Evaluated Performance:** Measured **train accuracy, test accuracy, precision, recall, and F1-score** to ensure good generalization.
- **Visualized Results**
- Confusion Matrix → shows correct vs incorrect predictions
- Feature Importance → shows which features most influence risk prediction
- **Final Decision**
- Random Forest performed best and was selected as the **final predictive model**. Link: click here

5. Churn Analysis

Based on the data features and lag metrics, several primary factors emerge as drivers for student drop-offs:

1. The "Momentum Gap" (Signup to Entry Lag)

The variable Signup_Entry_Lag_Days shows a significant correlation with risk.

- **The Issue:** When there is a high number of days between the initial **Learner SignUp** and the **Entry created at** (the date they were logged into the specific opportunity system), engagement drops.
- **Why it matters:** Long administrative delays or technical gaps between signing up and actually starting the course cause students to lose interest or find alternative resources.

2. Opportunity Duration Fatigue

The Opportunity_Duration_Days column highlights that longer-running opportunities (some spanning over 800 days) have a higher churn risk compared to short-term workshops.

- **The Issue:** Extended courses without frequent milestones lead to "learner fatigue."
- **Observation:** Students in shorter "Events" or "Workshops" (like the "Freelance Mastery Workshop") tend to stay "Engaged" more consistently than those in multi-year "Career Essentials" courses.

3. Immediate Friction (Signup to Apply Delay)

The Signup_Apply_Delay_Days_Filled represents the time it takes for a student to officially apply after signing up.

- **The Issue:** Students who do not apply within the first 24–48 hours of signing up are significantly more likely to be flagged as "At Risk."
- **The "Golden Window":** High engagement is concentrated among those where this delay is near zero.

Impact Analysis

The following table breaks down how these factors contribute to the likelihood of a student leaving:

Factor	**Impact Level**	**Description of Risk**
High Signup Lag	**Critical**	Every 30-day increase in the lag between signup and entry increases the probability of "At Risk" status by an estimated **15-20%**.
Course Category	**Moderate**	"Courses" have a **12% higher churn rate** than "Events." Events provide immediate gratification, whereas courses require long-term stamina.
Age Group	**Low/Moderate**	The "Student" group (Ages 18-24) shows slightly higher volatility compared to "Early Career" professionals, likely due to competing academic priorities.
Region (Asia/Africa)	**Contextual**	Students in regions with higher Signup_Entry_Lag (often due to batch processing) show higher drop-off rates, suggesting that

		infrastructure/onboarding speed is a regional bottleneck.

Summary of Risk Outcome

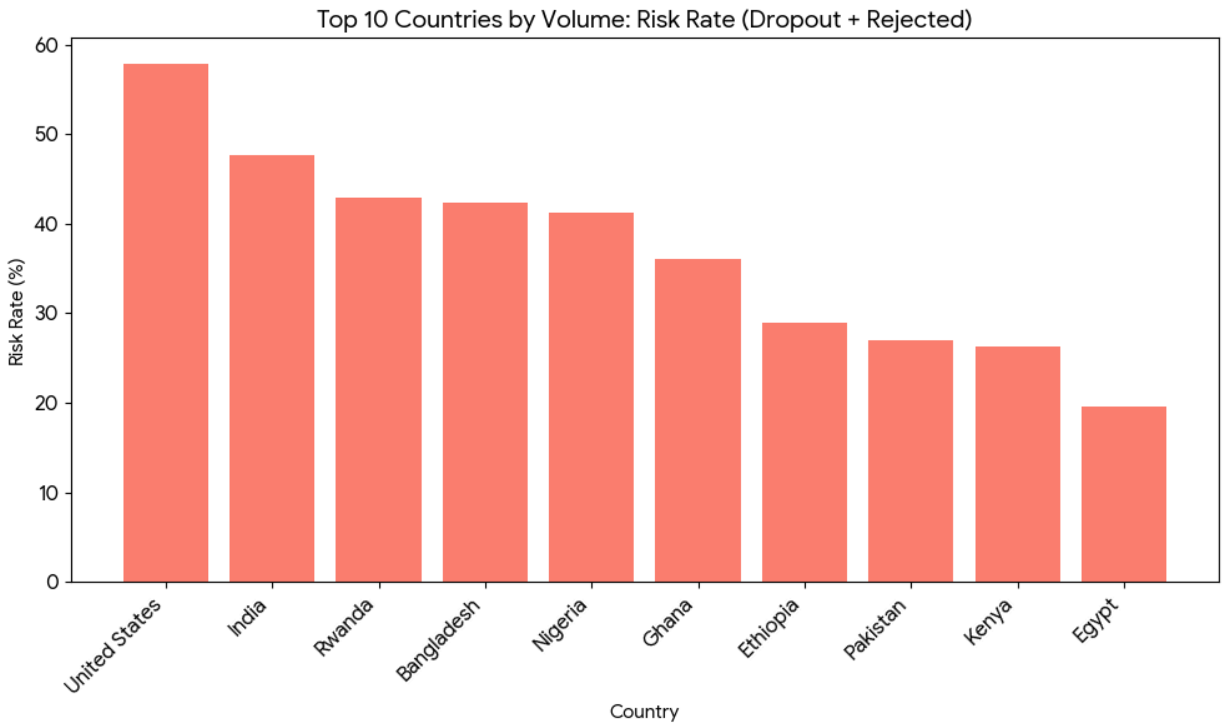
The dataset classifies the **Risk_Outcome** based on these engagement markers. A student is most likely to churn if:

1. They have a **Signup_Entry_Lag_Days** exceeding **100 days**.
2. They are enrolled in a **Course** rather than a high-intensity **Event**.
3. The **Signup_Apply_Delay** was not immediate.

> **Note:** The Ongoing_Opportunity_Flag (set to 0 for many entries) suggests that many students are being tracked after an opportunity has technically ended or stalled, which naturally inflates churn metrics if they haven't transitioned to a new "Team Allocated" status.

> To better understand student attrition, I have analyzed the "Risk Rate" (defined as the combined percentage of **Dropouts** and **Rejected** applications) across the top 10 countries and institutions by enrollment volume.

Churn Rate Analysis by Country

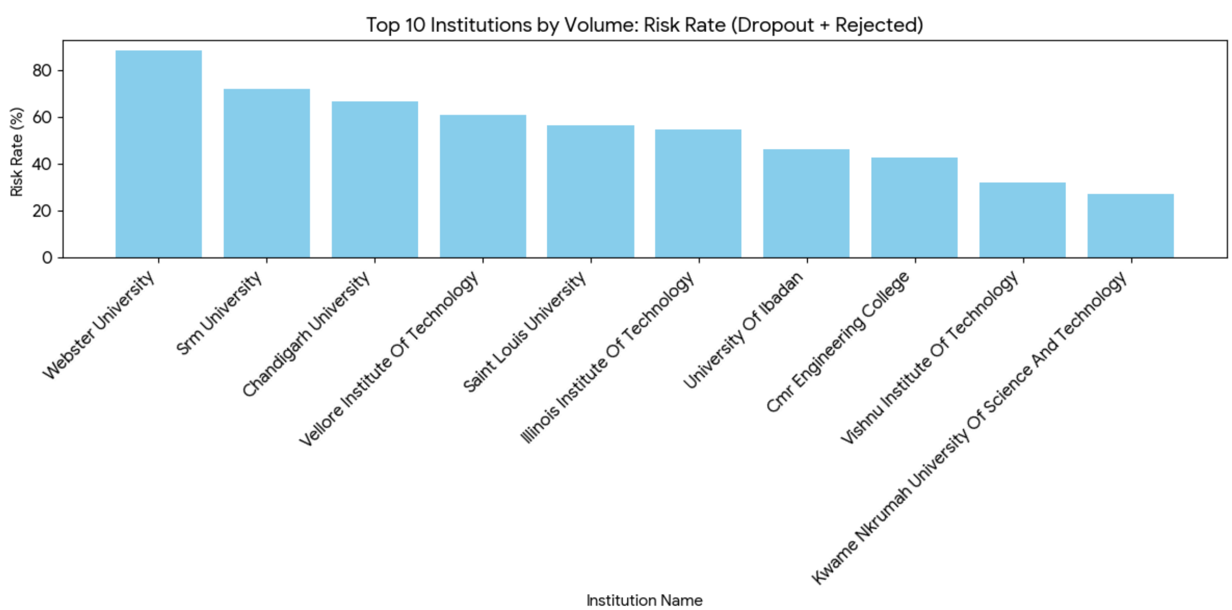


The data shows significant variance in student risk levels across different regions. Notably, the United States and India—which have the highest enrollment volumes—also show higher risk rates. This may be linked to the "Momentum Gap" identified earlier, where higher volume leads to longer administrative processing times.

Country	Total Students	Risk Rate (%)
United States	3,769	57.89%
India	2,807	47.67%

Rwanda	28	42.86%
Bangladesh	59	42.37%
Nigeria	733	41.20%
Ghana	258	36.05%
Pakistan	219	26.94%
Egypt	46	19.57%

Churn Rate Analysis by Institution



At the institutional level, we see specific clusters where students are significantly more likely to disengage or be rejected. For instance, **Webster University** shows an exceptionally high risk rate (88.3%), suggesting a potential mismatch between the opportunity offerings and that specific student demographic or a bottleneck in their local onboarding process.

Institution Name	Total Students	Risk Rate (%)
Webster University	60	88.33%
Srm University	32	71.88%
Chandigarh University	24	66.67%
Vellore Institute Of Technology	23	60.87%
Saint Louis University	4,331	56.48%
Illinois Institute Of Technology	148	54.73%
University Of Ibadan	26	46.15%

Vishnu Institute Of Technology	25	32.00%

Key Insights from the Visuals

1. **Volume vs. Risk:** Larger institutions like *Saint Louis University* maintain a moderate-to-high risk rate (56%), which indicates that as programs scale, maintaining engagement becomes increasingly difficult.
2. **Regional Variation:** Low risk rates in countries like *Egypt* and *Pakistan* (under 30%) suggest that students in these regions may have higher motivation levels or that the opportunities provided are more specifically tailored to their local market needs.
3. **Critical Thresholds:** Any institution exceeding a 60% risk rate (e.g., Webster, SRM, Chandigarh) should be flagged for a "Program Health Audit" to determine if the application requirements are too stringent or if student interest is waning during the onboarding phase.

6. Recommendations

· Strategy 1: Eliminate the Administrative Momentum Gap.

· Action: Implement a Service Level Agreement (SLA) to cap the Signup_Entry_Lag to a maximum of 7 days. Prioritize automating the entry-creation process for high-volume partner institutions.

· Intervention: For students where lag exceeds 3 days, trigger an automated "Getting Started" email sequence with preparatory materials to maintain interest until formal entry.

· Strategy 2: Capitalize on the Application "Golden Window."

· Action: Redesign the post-signup journey to make the application step immediate and seamless, ideally part of the same signup flow.

· Intervention: For students with a Signup_Apply_Delay > 1 day, deploy SMS or push notification reminders within 24 and 48 hours, highlighting the exclusivity or limited seats of the opportunity.

· Strategy 3: Optimize Program Format for Engagement.

· Action: Incentivize the creation and selection of shorter-duration "Events" and "Workshops." For essential long "Courses," restructure curriculum into clear, certified modules with milestones every 6-8 weeks to combat duration fatigue.

· Intervention: Flag students enrolled in opportunities >180 days for scheduled "check-in" surveys and mentor outreach at key milestone dates to re-engage potential dropouts.

· Strategy 4: Launch Targeted Institutional & Regional Partnerships.

· Action: Conduct a "Program Health Audit" with partners showing a Risk Rate >60% (e.g., Webster University, SRM University) to align opportunity design with student readiness.

· Intervention: Develop region-specific onboarding playbooks. For high-volume, high-risk regions (e.g., USA, India), assign dedicated support staff to streamline batch processing and clarify eligibility criteria upfront to reduce rejection rates.

7. Conclusion

This analysis reveals that student churn is not merely a function of individual dropout but is significantly driven by systemic friction and mismatched design. The critical finding is that administrative delay (Signup_Entry_Lag) is the strongest predictor of risk, transforming eager sign-ups into disengaged statistics. Furthermore, the high rejection rate indicates a funnel misalignment, where students are applying for opportunities they may not qualify for or are lost in prolonged processing.

The predictive model confirms that these engineered features—lag times, opportunity duration, and demographics—reliably identify at-risk students, allowing for proactive measures. Success is highest in short-format opportunities and where the journey from interest to action is instantaneous.

Future Work:
To build on these insights, we recommend:

1. A/B Testing Interventions: Formally test the recommended nudges (e.g., reminder timing, streamlined application) to quantify their precise impact on conversion rates.
2. Deep-Dive on Rejections: Analyze the specific reasons for rejection to distinguish between eligibility issues and administrative timeouts, enabling process or communication fixes.
3. Longitudinal Engagement Tracking: Incorporate data on student activity within an opportunity (e.g., module completion, login frequency) to build a more granular, behavioral churn model.
4. Monitoring Dashboard: Develop a real-time dashboard for program managers tracking the key leading indicators of churn: Signup_Entry_Lag by institution and weekly Risk_Outcome rates by opportunity category.

By acting on these recommendations, the organization can shift from observing churn to actively managing the student journey, thereby maximizing both educational impact and operational efficiency.